EXPERIMENT - 2

AIM: Convert the ER diagram for the Indian E-commerce platform into a Relational Schema. Write complete MYSQL CREATE TABLE statements with Primary Keys, Foreign Keys, NOT NULL, UNIQUE, ON DELETE CASCADE and ON DELETE, SET NULL. Insert Sample Data and Demonstrate referential integrity violation.

1.) Relational Schema:

Mapping Notes:

- Name (composite): First_Name, Last_Name in Customer
- Phone (Multi-valued, CUSTOMER only) → CUSTOMER_PHONE
- OrderItem is a weak Entity Identified by (Order ID, Product ID)
- SELLS is an M:N relationship between PRODUCT and SELLER → Junction table PRODUCT_SELLER.
- The ISA specialization of PRODUCT (Eletronics / Clothing / Grocery) is mapped using one table per subtype, PK = FK = Product ID

Table	Primary key	Important Foreign key
CUSTOMER	Customer_ID	—
CUSTOMER_Phone	(Customer ID, Phone)	CustomerID → Customer
ADDRESS	Address ID	Customer ID → Customer
SELLER	Seller ID	—
CATEGORY	CategoryID	—
PRODUCT	PRODUCT_ID	Category ID → CATEGORY.
PRODUCT_SELLER	(Product ID, Seller ID)	Product ID → PRODUCT, SellerID → Seller
ELECTRONICS	Product ID	Product ID → PRODUCT
CLOTHING	Product ID	Product ID → PRODUCT.
GROCERY	Product ID	Product ID → PRODUCT
ORDERS	Order ID	Customer ID → CUSTOMER
PAYMENT	Payment ID	Order ID → ORDERS
DELIVERY	Delivery ID	ORder ID → ORDERS

~~2. SQL Queries~~

2. SQL Queries

2.1 Schema Creation

```
DROP DATABASE IF EXISTS ECOMMERCE_ER;
CREATE DATABASE ECOMMERCE-ER;
USE ECOMMERCE_ER;
--1. CUSTOMER ( Name is a composite attribute → First_
   Name, Last_Name)
Create table CUSTOMER (
    CustomerID INT PRIMARY KEY,
    First_Name VARCHAR(30) NOT NULL,
    Last_Name VARCHAR(30) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    Date_of_Birth DATE NOT NULL.
);
--Phone is a multi-valued attribute of CUSTOMER →
  separate table.
Create table CUSTOMER_PHONE(
    CustomerID INT,
    Phone VARCHAR(15),
    PRIMARY KEY (CustomerID, Phone),
    Foreign Key(CustomerID) REFERENCES CUSTOMER(CustomerI
        ON DELETE CASCADE
);
--2. ADDRESS (CUSTOMER 1 -- HAS -- N Address, Address has
  total participation).
```

```
Create table ADDRESS(
    AddressID INT PRIMARY KEY,
    House_No  VARCHAR(10),
    Street    VARCHAR(50),
    City      VARCHAR(30) NOT NULL,
    State     VARCHAR(30) NOT NULL,
    PIN_Code  VARCHAR(10) NOT NULL,
    Address_Type VARCHAR(20),
    CustomerID INT NOT NULL,
    Foreign key(CustomerID) REFERENCES CUSTOMER(Customer
    ID)  ON DELETE CASCADE
);

--3. SELLER

Create table SELLER(
    SellerID INT PRIMARY KEY,
    Seller_Name VARCHAR(50) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    Phone VARCHAR(15),
    GSTIN VARCHAR(20) NOT NULL UNIQUE
);

--4. CATEGORY.

Create table CATEGORY(
    CategoryID INT PRIMARY KEY,
    CategoryName VARCHAR(50) NOT NULL UNIQUE,
    Description VARCHAR(150)
);
```

--5. PRODUCT (BELONGS_TO → Category ID, nullable so SET NULL can be shown)

```
Create table PRODUCT(
    ProductID INT PRIMARY KEY,
    Product_Name VARCHAR(100) NOT NULL,
    Price DECIMAL (10,2) NOT NULL,
    Description VARCHAR (200),
    Stock_Quantity INT NOT NULL DEFAULT 0,
    Brand VARCHAR(50),
    CategoryID INT,
    Foreign key (CategoryID) REFERENCES CATEGORY(CategoryID)
        ON DELETE SET NULL,
);
```

--SELLS is an M:N relationship between PRODUCT and SELLER → junction table.

```
Create table Product_Seller(
    ProductID INT,
    SellerID INT,
    Primary Key (ProductID, SellerID),
    Foreign Key (ProductID) References Product(Product_ID)
        ON DELETE CASCADE,
    Foreign key (SellerID) References Seller (SellerID)
        ON DELETE CASCADE
);
```

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--6. ISA Specialization of Product → Electronics/Clothing/Grocery

```
Create table Electronics(
    ProductID INT PRIMARY KEY,
    Warranty_Period VARCHAR(30),
    Model_Number VARCHAR(50),
    Foreign key (ProductID) References Product(ProductID)
        ON DELETE CASCADE
);
Create table Clothing(
    ProductID INT PRIMARY KEY,
    Size VARCHAR(10),
    Color VARCHAR(30),
    Material VARCHAR(30),
    Foreign key (ProductID) references Product(ProductID)
        ON DELETE CASCADE
);
Create table Grocery(
    ProductID INT PRIMARY KEY,
    OrderDate Date NOT NULL,
    OrderStatus VARCHAR(20) NOT NULL,
    TotalAmount DECIMAL(10,2) NOT NULL,
    CustomerID INT NOT NULL,
    Foreign key (CustomerID) references Customer(CustomerID)
        ON DELETE CASCADE
);
Create table Orders(
    OrderID INT Primary Key,
    OrderDate DATE NOT NULL,
    Order_status VARCHAR(20) NOT NULL,
    CustomerID INT NOT NULL,
```

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```
    Foreign key (customerID) References customer (customerID)
        ON DELETE CASCADE
    );
Create table Payment(
    PaymentID INT Primary key,
    Payment_Date DATE NOT NULL,
    Amount DECIMAL(10,2) NOT NULL,
    Payment_method VARCHAR(30) NOT NULL,
    TransactionID VARCHAR(30) UNIQUE,
    Foreign key (OrderID) references ORDERS (orderID)
        ON DELETE CASCADE
    );
-- Delivery (Delivered_By: order 1 --- 1 DELIVERY)
Create table Delivery(
    DeliveryID INT PRIMARY KEY,
    Delivery_Date DATE,
    Delivery_status VARCHAR(20) NOT NULL,
    TrackingID VARCHAR(30) UNIQUE,
    Delivery_Partner VARCHAR(50),
    OrderID INT NOT NULL UNIQUE,
    Foreign key (OrderID) references ORDERS (orderID)
        ON DELETE CASCADE
    );
```

2.2 Sample data (Insert)

```
USE ECOMMERCE_ER;
Insert into customer values
(1, 'Kunal', 'Chaudhary', 'email7@gmail.com', '2006-01-01');
```

Insert into Customer_Phone Values
('1', '9876543210'),
('1', '0123456789');

Insert into ADDRESS Values
('113', 'A', 'XYZ City', 'Lucknow', 'Uttar Pradesh', '****', 'Home', 1);

Insert into SELLER Values
('201', 'Electronics', 'Electronic gadget and accessories');

Insert into Product Values
('301', 'Gaming laptop', 75000.00, '15-6 inch, RTX Graphics', '10', 'ASUS', '201');

Insert into Electronics Values
('301', '2 Year', 'ASUS TUF-F15');

Insert into Product_seller values
('301', '101');

Insert into Payment Values
('601', '2026-08-14', 75000.00, 'UPI', 'Paid', 'TXN123', 501);

Insert into Delivery values
('701', NULL, 'Processing', 'TRK123', 'Delivery', 501);

2.3 Referential Integrity Violation

Insert into order values
('502', '2026-08-14', 'Placed', 5000.00, 999);

2.4 ON DELETE SET NULL DEMONSTRATION

Delete from CATEGORY WHERE Category ID = 201;

2.5. ON DELETE CASCADE demonstration

Delete from Customers where CustomerID = 1;

* Output -

- **Output 1 - SHOW TABLES**

```
SHOW TABLES;
```

Tables_in_ECOMMERCE_ER
Address
Category
Clothing
Customer
Customer_Phone
Delivery
Electronic
Grocery
Orders
Payment
Product
Product_Seller
Seller

- **Output 2 - Sample Data**

```
Select * from Customers;
```

CustomerID	First_Name	Last_Name	Email	Date of Birth
1	Kunal	Chaudhary	email2@gmail.com	2006-01-10

```
select * from Product;
```

ProductID	ProductName	Price	stockQty	Brand	CategoryID
301	Gaming laptop	75000.00	10	ASUS	201

SELECT * from orders;

OrderID	Order_Date	Order_Status	Total_Amount	CustomerID
501	2026-08-14	Placed	75000.00	1

- Output 3 - Referential integrity violation

Error 1452(23000): Cannot add or update a child row: a foreign key constraint failed ('Ecommerce_ER'.'Orders', CONSTRAINT 'ORDER_ibfk_1')
Foreign Key ('CustomerID') references 'Customer' ('CustomerID') ON DELETE CASCADE

- Output 4 - ON DELETE SET NULL (category - PRODUCT)

Before deletion:

ProductID	ProductName	CategoryID
301	Gaming laptop	201

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After deletion:

Product ID	ProductName	CategoryID
301	Gaming laptop	NULL

- Output 5 - ON DELETE CASCADE (customer → everything downstream)

Table	Row count after DELETE
ORDERS	0
ORDERITEM	0
PAYMENT	0
DELIVERY	0
ADDRESS	0
CUSTOMER_PHONE	0

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